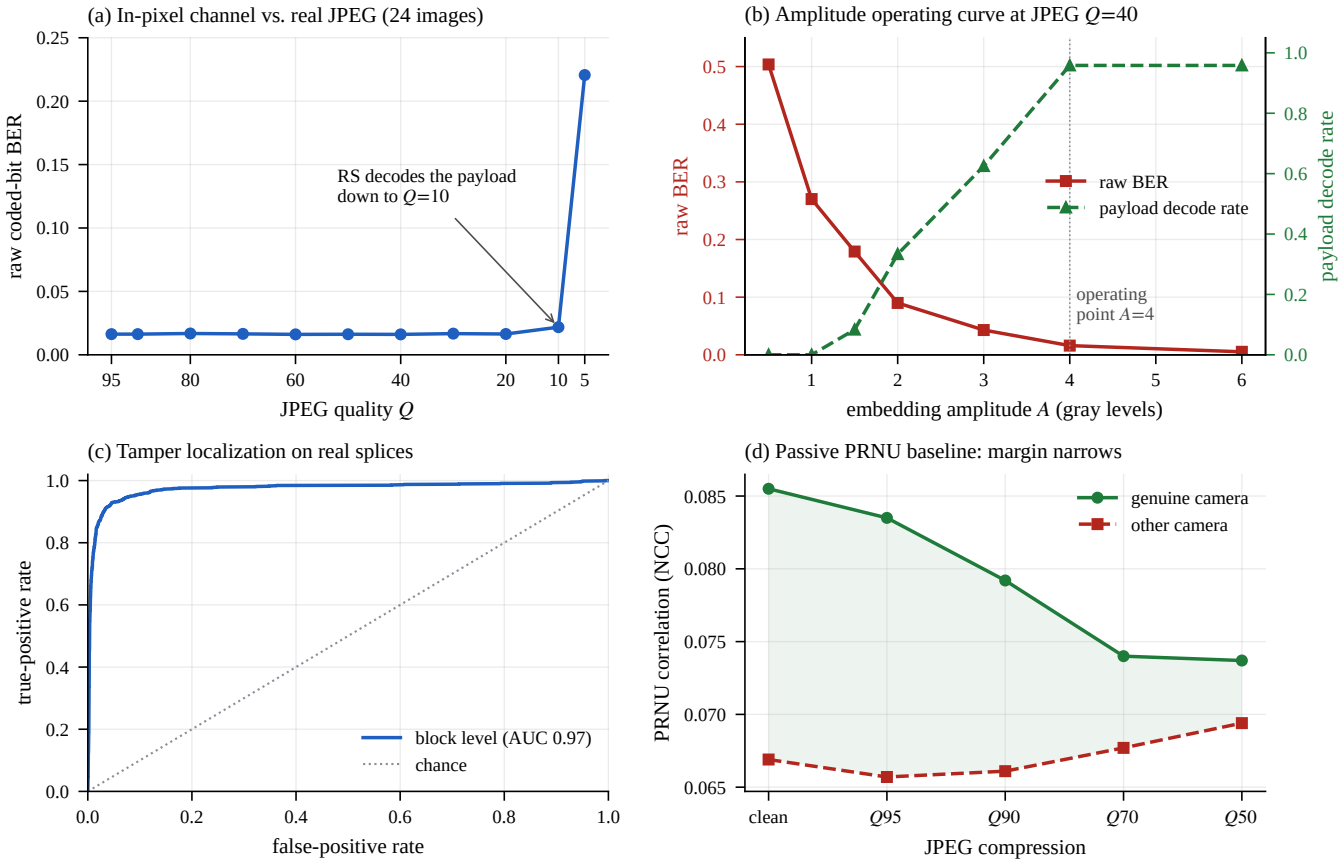
(a) In-pixel channel vs. real JPEG (24 images)
raw coded-bit BER
0.25
0.20
0.15
0.10
0.05
0.00
RS decodes the payload
down to Q=10
95
80
60
40
20
10
5
JPEG quality Q
(b) Amplitude operating curve at JPEG Q=40
raw BER
0.5
0.4
0.3
0.2
0.1
0.0
payload decode rate
1.0
0.8
0.6
0.4
0.2
0.0
raw BER
payload decode rate
operating
point A=4
1
2
3
4
5
6
embedding amplitude A (gray levels)
(c) Tamper localization on real splices
true-positive rate
1.0
0.8
0.6
0.4
0.2
0.0
block level (AUC 0.97)
chance
0.0
0.2
0.4
0.6
0.8
1.0
false-positive rate
(d) Passive PRNU baseline: margin narrows
PRNU correlation (NCC)
0.085
0.080
0.075
0.070
0.065
genuine camera
other camera
clean
Q95
Q90
Q70
Q50
JPEG compression